

# ETH Z - Lin Alg HS 24

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Angle between two vectors:  $\cos(w) = \frac{v \cdot w}{\|v\| \|w\|}$

Cauchy-Schwarz inequality:  $|v \cdot w| \leq \|v\| \|w\|$

(equal if  $v$  is multiple of  $w$ )

Triangle Inequality:  $\|v + w\| \leq \|v\| + \|w\|$

Hyperplane:  $H = \{v \in \mathbb{R}^n \mid v \cdot d = 0\}$

Scalar Product:  $v \cdot w = v^T w = \langle v, w \rangle = \sum_{i=1}^n v_i w_i = v_1 w_1 + \dots$

Linear transformation:  $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$

(i)  $T(x+y) = T(x) + T(y)$

(ii)  $T(\lambda x) = \lambda T(x)$

Gaussian elimination: transform  $Ax=b$

to  $Ux=c$ , where the solution  $x$  stays the same

operations: 1) add multiples of a row to another row  
2) swap two rows

$$[A|b] \Rightarrow [U|c] \quad O(m^3)$$

Just works with lin indep. square  $A$

just for  $A \in \mathbb{R}^{n \times n}$

Inverse:  $MM^{-1} = M^{-1}M = I$

2x2:

$$M = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \Rightarrow M^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

general:

$$[A|I] \xrightarrow{\text{Gauss-Jordan}} [I|A^{-1}]$$

Inverse Theorem (T3.11):

(i)  $A$  is invertible

(ii) for every  $b \in \mathbb{R}^m$ ,  $Ax=b$  has unique solution

(iii) columns of  $A$  are linearly independent

LU-Decomposition (=LR):  $A=LU$

(L: lower triangular, U: upper triangular)

U = product of Gauss (no permutations)

$$L = \begin{bmatrix} 1 & 0 & 0 \\ c_{21} & 1 & 0 \\ c_{31} & c_{32} & 1 \end{bmatrix} \leftarrow \text{keep track while doing Gauss}$$

We subtracted  $c_{31}$ -times  $R_1$  from  $R_3$

solving  $Ax=b$  with  $A=LU$ :  $LUx=b$

- solve  $Ly=b$  with forward substitution

- solve  $Ux=y$  with back substitution

LUP-Decomposition:  $PA=LU$

same as LU but  $P$  is a permutation matrix

solving  $Ax=b$  with  $PA=LU$ :  $P^T L U x = b$

- solve  $P^T z$  for  $z$

- same as LU

Gauss-Jordan elimination: transform

$A$  to  $R_0 = MA$ , where  $x$  is the same

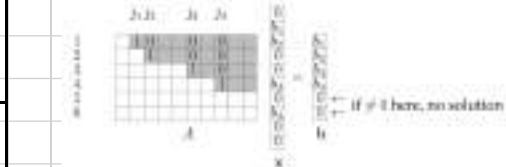
( $R$  is in REF or RREF)

$R$  in REF  $[i_1, \dots, i_r]$  has rank  $r$  (D3.20)

$A$  has independent columns  $j_1, \dots, j_r$

Like Gauss, but we also eliminate above the pivot

get  $M$ : transform  $m \times m$   $I$  with same row ops  
solve  $Ax=b$  with  $R_0$  (or given  $M$  in  $O(m^2)$ )



CR decomposition:  $A = C R$

$C$ : independent columns of  $A$ , columns  $j_1, \dots, j_r$  in REF

$R$ : RREF from Gauss-Jordan elimination

Vector spaces: Triple  $(V, +, \cdot)$  (D4.1)

- $v + w = w + v$  (commutativity)
- $u + (v + w) = (u + v) + w$  (associativity)
- There is a vector  $0$  such that  $v + 0 = v$  for all  $v$  (zero vector)
- There is a vector  $-v$  such that  $v + (-v) = 0$  (negative vector)
- $1 \cdot v = v$  (identity element)
- $(\lambda \mu)v = \lambda(\mu v)$  (compatibility of  $\cdot$  and  $\cdot$ )
- $\lambda(v + w) = \lambda v + \lambda w$  (distributivity over  $+$ )
- $(\lambda + \mu)v = \lambda v + \mu v$  (distributivity over  $+$ )

Subspace (D4.8) non-empty  $U \subseteq V$  with:

(i)  $v + w \in U$

(ii)  $\lambda v \in U$

( $\Rightarrow 0 \in U$ )

$U$  is also a vectorspace (L4.12)

Basis:  $B \subseteq V$ , if  $B$  is linearly independent and  $\text{Span}(B) = V$

Fundamental subspaces:

Column space  $C(A)$ :

independent columns of  $A$  form basis (L4.17)

basis = {columns  $j_1, \dots, j_r$  in REF}, see CR

Row space  $R(A)$ :  $R(A) = C(A^T)$

independent rows of  $A$  form a basis

basis = {rows of RREF}, see CR

Nullspace  $N(A)$ : (= Ker(A))

set of all solutions  $Ax=0$

extract basis from  $R$  in RREF Example:

$$R = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -2 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -2 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -2 & 2 \end{bmatrix} - \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -2 \end{bmatrix} = 0$$

$x(I) = -R x(a)$ , free variables  $x(a)$  substituted with arbitrary real numbers ( $e_1, e_2$  make sense), basic variables  $x(I)$  then determined

$$\begin{array}{l} \text{free variables} \\ \text{basic variables} \\ \text{nullspace equation} \end{array} \quad \begin{array}{l} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -2x_3 - 3x_4 \\ x_3 \\ x_4 \\ 0 \end{bmatrix} \\ \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -2x_3 - 3x_4 \\ x_3 \\ x_4 \\ 0 \end{bmatrix} \end{array}$$

Left Nullspace  $LN(A)$ :  $LN(A) := N(A^T)$  (D4.36)

solutions of  $yA=0$

basis = {last  $m-r$  rows of  $M(R_0=MA)$ }

Solution Space  $\text{Sol}(A, b)$ :

$\text{Sol}(A, b) := \{x \in \mathbb{R}^n \mid Ax=b\}$ , if  $b \neq 0$   $\text{Sol}(A, b)$  is not subspace, since doesn't contain zero vector

$\text{Sol}(A, b) = \{s+x : x \in N(A)\}$  (T4.40)

$$\begin{array}{c} \text{LN(A)} \quad \text{C(A)} \quad \text{N(A)} \quad \text{R(A)} \\ \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -2 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -2 & 2 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -2 & 2 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -2 \end{bmatrix} \end{array}$$

Number of solutions:

| #sol.   | $r=n$                    | $r < n$                         |
|---------|--------------------------|---------------------------------|
| $r=m$   | Invertible<br>1          | Underdetermined<br>$\infty$     |
| $r < m$ | Overdetermined<br>0 or 1 | Overdetermined<br>0 or $\infty$ |

Steinitz exchange Lemma:  $V$  vector space,

$F \subseteq V$  set of linearly independent vectors,

$G \subseteq V$  finite set of vectors with  $\text{Span}(G) = V$

(i)  $|F| \leq |G|$

(ii) There exists  $E \subseteq G$  of size  $|G| - |F|$  such that  $\text{Span}(F \cup E) = V$

Dimension  $\dim(V)$  = size of any basis of  $V$

Orthogonality:  $v, w$  are orthogonal if

$v^T w = 0$ , subspaces  $V, W$  are orthogonal if all their vectors are (D5.11)

$V \perp W = 0$  (L5.14)

Orthogonal complement  $V^\perp = \{w \in \mathbb{R}^n \mid w^T v = 0 \text{ for all } v \in V\}$  (D5.15)

$N(A) = C(A^T)^\perp = R(A)^\perp$  (T5.16)

$(V^\perp)^\perp = V$  (L5.18)

**Projections:** projection of  $v$  to subspace  $S$  is closest point:  
 $\text{proj}_S(b) = \arg \min_{p \in S} \|b - p\|$

For a line  $S = \{ \lambda a \mid \lambda \in \mathbb{R} \} = C(a)$ :  
 $\text{proj}_S(b) = \frac{a \cdot b}{a \cdot a} a$  (LS.2.2)

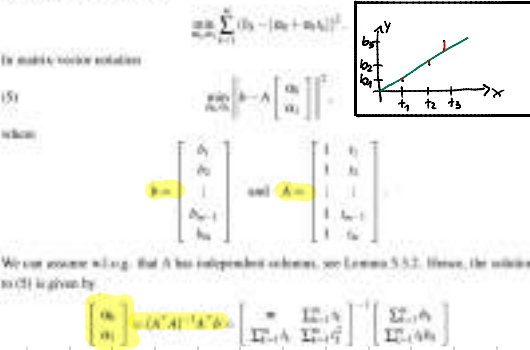
Generally,  $S = C(A)$ :  $\text{proj}_S = A \hat{x}$ ,  
 where  $\hat{x}$  satisfies the normal equations:  
 $A^T A \hat{x} = A^T b$  (LS.2.3)

If  $A$  has linearly independent columns,  
 $A^T A$  is square, invertible and symm. (CS.2.5)

columns of  $A$  are basis of  $S$ :  
 $\text{proj}_S(b) = P b$ ,  $P = A(A^T A)^{-1} A^T$

**Least Squares:**  $\min_{\hat{x} \in \mathbb{R}^n} \|A \hat{x} - b\|^2$   
 projection of  $b$  on  $C(A)$   
 normal equations:  $A^T A \hat{x} = A^T b$   
 When  $A$  has independent columns:  
 $\hat{x} = (A^T A)^{-1} A^T b$  (Fact 5.3.1)

**Linear Regression:**  $b \approx \alpha_0 + \alpha_1 t_k$



for a parabola:  $b_k = \alpha_0 + \alpha_1 t_k + \alpha_2 t_k^2$   
 $A = \begin{bmatrix} 1 & t_1 & t_1^2 \\ 1 & t_2 & t_2^2 \\ 1 & t_3 & t_3^2 \end{bmatrix}$

**Orthonormal Bases:**  
 $v_1, \dots, v_k$  are orthonormal if they are orthogonal and have norm 1 ( $\|v_i\| = 1$ ) (DS.4.1)

A square matrix  $Q$  is orthogonal if  $Q^T Q = I$ ,  $Q^{-1} = Q^T$ , the columns of  $Q$  form a basis of  $\mathbb{R}^n$  (columns are **orthonormal**)

$\|Qx\| = \|x\|$  and  $(Qx)^T (Qy) = x^T y$  (PS.4.6)

For least squares:  $\hat{x} = Q^T b$  (P.4.5.7) ( $Q^T Q = I$ )

**Gram-Schmidt Process:** (Algorithm 5.4.9)  
 Given linearly independent vectors  $a_1, \dots, a_n$ , Gram-Schmidt returns orthonormal basis of  $\text{Span}(a_1, \dots, a_n)$  (TS.4.10)

**QR-De composition:**  $A = QR$   
 $A$  needs to have linearly independent columns.  
 $Q$  is orthogonal (output of Gram-Schmidt)  
 $R = Q^T A$  (upper triangular and invertible) (LS.4.12)

With QR we have:  
 $C(A) = C(Q)$ ,  $\text{proj}_{C(A)} = QQ^T b$   
 Least squares:  $R \hat{x} = Q^T b$

**Pseudoinverse (Moore-Penrose Inverse):**  
 For **full column rank**:  $A^+ = (A^T A)^{-1} A^T$   
 (Left inverse of  $A$ :  $A^+ A = I$  (PS.5.2))  
 For **full row rank**:  $A^+ = A^T (A A^T)^{-1}$   
 (Right inverse of  $A$ :  $A A^+ = I$  (LS.5.4))  
 $\hat{x}$  minimizes:  $\min_{\hat{x}} \|x\|^2$ ,  $\hat{x} = A^+ b$ ,  $\hat{x} \in C(A^T)$

For all matrices:  $A^+ = R^+ C^+$   
 $= R^+ (C^T A R^T)^{-1} C^T$   
 What  $A^+$  achieves:

the (unique) solution to  $\min_{x \in \mathbb{R}^n} \|x\|^2$  s.t.  $A^T A x = A^T b$  is given by  $\hat{x} = A^+ b$

$\text{rank}(A) = r$ ,  $S \in \mathbb{R}^{m \times r}$ ,  $T \in \mathbb{R}^{r \times n}$ ,  $A = ST$ :  
 $A^+ = T^+ S^+$  (PS.5.9)

- (1)  $A A^+ A = A$ , (2)  $A^+ A A^+ = A^+$ , (3)  $A A^+$  is symmetric and projection matrix of  $C(A)$
- (4)  $A^+ A$  is symmetric and projection matrix of  $C(A^T)$
- (5)  $(A^T)^+ = (A^+)^T$  (TS.5.11)

**Determinant:** How much does the linear transformation of  $A$  transform space

just works on square matrices

**2x2:**  $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$   
**3x3:**

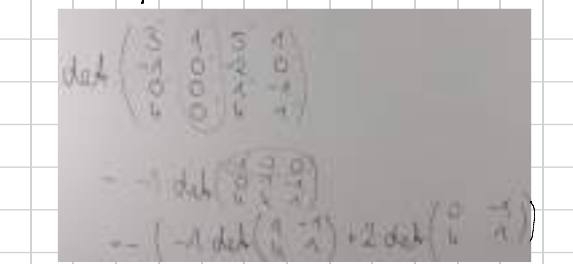
**n x n:** sign of a permutation:  
 $\text{sgn}(\sigma) = \begin{cases} 1 & \text{even number of swaps} \\ -1 & \text{odd number of swaps} \end{cases}$

$\det(A) = \sum_{\sigma \in \Pi_n} \text{sgn}(\sigma) \prod_{i=1}^n A_{i, \sigma(i)}$ , where  $\Pi$  is the set of all permutations of  $n$  elements ( $|\Pi| = n!$ )

for permutation matrices:  $\det(P) = \text{sgn}(P)$   
 for triangular matrices:  $\det(T) = \prod_{k=1}^n t_{kk}$   
 for orthogonal matrices:  $\det(Q) = 1$  or  $-1$   
 $\det(I) = 1$ ,  $\det(A^T) = \det(A)$

$A$  is invertible iff  $\det(A) \neq 0$  (PG.0.11)  
 $\det(AB) = \det(A) \det(B)$   
 $\det(A^{-1}) = \frac{1}{\det(A)}$

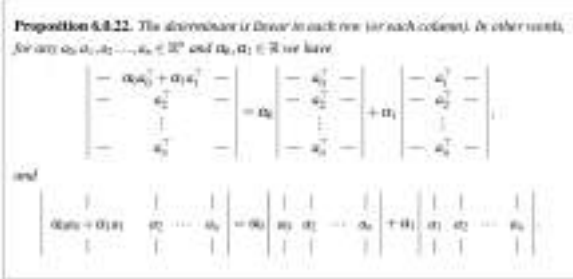
**cofactor expansion (Laplace):** calc det of  $n \times n$   
 $C_{ij} = (-1)^{i+j} \det(A_{ij})$ ,  $A_{(n-1) \times (n-1)}$  obtained by removing row  $i$  column  $j$   
 for any  $i$ :  $\det(A) = \sum_{j=1}^n A_{ij} C_{ij}$   
 (works for columns as well but not mentioned in the script)



$A^{-1} = \frac{1}{\det(A)} C^T$ ,  $C$  co-factors of  $A$  as entries (PG.0.17)  $\leftarrow$  computationally bad!

**Carmers rule:** solve  $Ax = b$  (inefficiently)  
 $X_j = \frac{\det(A_{Bj})}{\det(A)}$ ,  $A_{Bj}$  is matrix obtained by  $A$  by replacing  $j$ -th column with  $b$

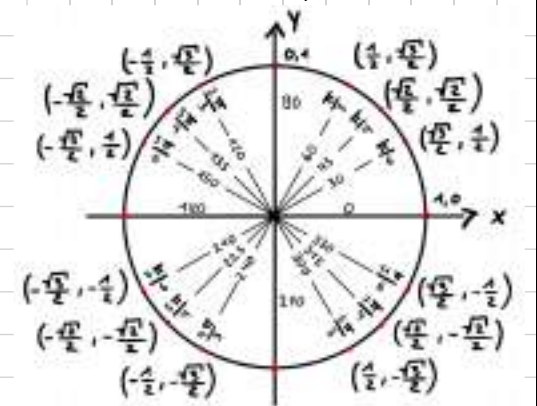
$\det(A) = \frac{1}{\det(P)} \det(L) \det(U) = \text{sgn}(P) \det(U)$



**Complex Numbers:**  
 Given  $z \in \mathbb{C}$  with  $z = a + ib$  we have the following notation  
 (15)  $\Re(a + ib) = a$  called the real part of  $z = a + ib$ ,  
 (16)  $\Im(a + ib) = b$  called the imaginary part of  $z = a + ib$ ,  
 (17)  $|z| = \sqrt{a^2 + b^2}$  called the modulus of  $z = a + ib$ ,  
 (18)  $\bar{a + ib} = a - ib$  called the complex conjugate of  $z = a + ib$ .

**Euler's formula:**  $e^{i\theta} = \cos \theta + i \sin \theta$   
 $\Rightarrow e^{i\pi} = -1$  often written  $e^{i\pi} + 1 = 0$

**Polar Coordinates:**  $z \in \mathbb{C}$  can be expressed as  $z = re^{i\theta}$ , where  $r \geq 0$  is the modulus of  $z$  and  $\theta \in \mathbb{R}$  ( $\theta \in [0, 2\pi]$ ) is an angle, called argument of  $z$



**Theorem 7.1.3 (Fundamental Theorem of Algebra):** Any degree  $n$  non-constant ( $n \geq 1$ ) polynomial  $P(z) = a_n z^n + a_{n-1} z^{n-1} + \dots + a_1 z + a_0$  with  $a_n \neq 0$  has at least one zero.  $\lambda \in \mathbb{C}$  such that  $P(\lambda) = 0$ .

$P(z) = a_n (z - \lambda_1)(z - \lambda_2) \dots (z - \lambda_n)$ , number of times a  $\lambda \in \mathbb{C}$  appears in this is called **algebraic multiplicity** of the zero (C7.0.4)

**Complex vectors/matrices:**  
"conjugate transpose" or "hermitian transpose"  $A^* = \bar{A}^T$   
inner product (= dot-product):  
 $\|v\|^2 = v^* v = \bar{v}^T v = \sum_{i=1}^n \bar{v}_i v_i = \sum_{i=1}^n |v_i|^2$

**Eigenvalues and Eigenvectors:**  
 $Av = \lambda v$ ,  $\lambda \in \mathbb{C}$  is eigenvalue of  $A$ ,  
 $v \in \mathbb{C}^n \setminus \{0\}$  is eigenvector of  $A$   
If  $\lambda \in \mathbb{R}$ , we call it real eigenvalue

$\lambda$  is eigenvalue of  $A$  iff  $\det(A - \lambda I) = 0$ ,  $v$  eigenvector associated with  $\lambda$  iff non-zero element of  $N(A - \lambda I)$

calc  $\lambda$  by solving formula  $\det(A - \lambda I) = 0$  for  $\lambda$ , then solve  $(A - \lambda I)v = 0$  for every  $\lambda$

$\det(A - \lambda I)$  is polynomial in  $\lambda$  (P7.1.3)  
Fundamental Theorem of Algebra implies: Every  $A \in \mathbb{R}^{n \times n}$  has eigenvalues (in  $\mathbb{C}$ )

$\lambda, v$  eigenvalue-eigenvector pair of  $A$ ,  
 $\lambda^k$  and  $v$  eigenvalue-eigenvector pair of  $A^k$  (P7.1.6)

$\lambda, v$  eigenvalue-eigenvector pair of invertible  $A$ ,  
 $\frac{1}{\lambda}, v$  eigenvalue-eigenvector pair of  $A^{-1}$  (P7.1.7)

If  $\lambda_1, \dots, \lambda_k$  are distinct, corresponding  $v_1, \dots, v_k$  are linearly independent (P7.1.8)

$A \in \mathbb{R}^{n \times n}$  with  $n$  distinct real eigenvalues, then  $v_1, \dots, v_n$  is basis made up of eigenvectors of  $A$  (T7.1.9), we say  $A$  has a complete set of real eigenvalues (D7.1.20)

eigenvalues of  $A$  are eigenvalues of  $A^T$  vectors not

**Trace:**  $\text{Tr}(A) = \sum_{i=1}^n A_{ii}$

(i)  $\text{Tr}(AB) = \text{Tr}(BA)$ , (ii)  $\text{Tr}(ABC) = \text{Tr}(BCA) = \text{Tr}(CAB)$  (L7.1.14)

$\lambda_1, \dots, \lambda_n$  the  $n$  eigenvalues of  $A \in \mathbb{R}^{n \times n}$   
( $\lambda$  may be repeated):  $\text{Tr}(A) = \sum_{i=1}^n \lambda_i$ ,  $\det(A) = \prod_{i=1}^n \lambda_i$   
helpful to find all  $\lambda$

**caution!**  
eigenvalues of  $A+B$  /  $AB$  not easily computed from those of  $A$  and  $B$   
- Gauss / LU don't preserve eigenvalues/vectors

$\lambda \in \mathbb{C}$  eigenvalue of orthogonal matrix, then  $|\lambda| = 1$  (P7.1.17)

projection matrix  $P$  has eigenvalues 0 and 1 and a complete set of eigenvectors (P7.1.21)

**geometric multiplicity** of  $\lambda$ : dimension of  $N(A - \lambda I)$

upper/lower triangular matrices have eigenvalues  $\lambda_i = t_{ii}$   $\begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} / \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}$

**Change of basis:** linear transformation  $L$  that takes vector  $x$  expressed by basis  $U$  and outputs the same vector  $x$  expressed by basis  $V$  ( $\beta$ )

$U$ : columns are basis  $u_1, \dots, u_n$  of  $\mathbb{R}^n$   
 $V$ : columns are basis  $v_1, \dots, v_m$  of  $\mathbb{R}^m$   
 $x = U \cdot \alpha = \sum_{j=1}^n \alpha_j u_j = U \cdot \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix}$

$x = V \cdot \beta = \sum_{j=1}^m \beta_j v_j = V \cdot \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_m \end{bmatrix}$   
 $\beta = \underbrace{V^{-1} U}_{\text{to basis } V} \alpha$ ,  $\alpha = \underbrace{U^{-1} V}_{\text{to basis } U} \beta$

**intuition:**  
 $x$  in canonical basis  $\xrightarrow{U^{-1} x} x_U$  expressed in basis  $U$   
 $x_U \xrightarrow{U x} x$  in canonical basis

**With linear transformation:**  
 $A$ : linear transformation  $L$  written in canonical basis

(1),  $L(x) = V \cdot \beta = \sum_{j=1}^n \beta_j v_j = V \cdot \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_n \end{bmatrix}$   
 $\beta = B \alpha$ , where  $B = \underbrace{V^{-1} A U}_{\text{linear transformation in canonical basis}}$

**Diagonalization:**

**Eigendecomposition:**

$A \in \mathbb{R}^{n \times n}$  with complete set of real eigenvectors  $v_1, \dots, v_n$ ,  $V = [v_1 \ v_2 \ \dots \ v_n] \in \mathbb{R}^{n \times n}$ ,  $\lambda_1, \dots, \lambda_n$  eigenvalues ( $\lambda_i$  associated to  $v_i$ ),  $\Lambda = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}$  ( $\Lambda_{ii} = \lambda_i, 0$  for  $i \neq j$ )  
 $A = V \Lambda V^{-1}$  (T7.2.1)  $A^k = V \Lambda^k V^{-1}$

**Diagonalizable Matrix:** if invertible  $V$  exists, such that  $V^{-1} A V = \Lambda$ , where  $\Lambda$  is diagonal

**Similar matrices:**  $A, B \in \mathbb{R}^{n \times n}$  are similar if there exists an invertible  $S$  such that  $B = S^{-1} A S$

**Spectral Theorem:** Any symmetric matrix  $A \in \mathbb{R}^{n \times n}$  has  $n$  real eigenvalues and an orthonormal basis made of eigenvectors of  $A$  (T7.3.1)

For all symmetric matrices exist orthogonal  $V$ , such that  $A = V \Lambda V^T$  (L7.3.2)

**Symmetric Matrices:**

The rank of a real symmetric matrix is the number of non-zero eigenvalues (counting repetitions) (L7.3.4)

$A \in \mathbb{R}^{n \times n}$  symmetric,  $\lambda_i \in \mathbb{R}$  (P7.3.7)

Every symm.  $A \in \mathbb{R}^{n \times n}$  has a real  $\lambda$  (L7.3.8)

**Rayleigh Quotient:** Given symm.  $A \in \mathbb{R}^{n \times n}$ , for  $x \in \mathbb{R}^n \setminus \{0\}$

$R(x) = \frac{x^T A x}{x^T x}$   
 $R(v_{\max}) = \lambda_{\max}$ ,  $R(v_{\min}) = \lambda_{\min}$ , where  $\lambda_{\min} / \lambda_{\max}$  is the smallest / biggest eigenvalue of  $A$  and  $v_{\min} / v_{\max}$  it's associated eigenvector



**Positive (semi)definit PSD/PD:**  
 Symmetric  $A \in \mathbb{R}^{n \times n}$  is:  
 - PSD if for all it's eigenvalues  $\lambda \geq 0$   
 - PD if for all it's eigenvalues  $\lambda > 0$   
 Symm  $A$  is:  
 - PSD iff  $x^T A x \geq 0$  for all  $x \in \mathbb{R}^n$   
 - PD iff  $x^T A x > 0$  for all  $x \in \mathbb{R}^n \setminus \{0\}$   
 (P7.3.12)

**Gram Matrix:** Given  $v_1, \dots, v_n \in \mathbb{R}^n$   
 $G_{ij} = v_i^T v_j$   
 if  $v_1, \dots, v_n$  columns of  $V$   $G = V^T V$

Given  $A \in \mathbb{R}^{m \times n}$  non-zero eigenvalues of  $A^T A$  and  $A A^T$  are the same,  $A^T A$  and  $A A^T$  are symmetric and PSD (P7.3.16)

**Cholesky decomposition:** Every symmetric PSD matrix  $M$  is a gram matrix of an upper triangular matrix  $C$   $M = C^T C$  (P7.3.17)

**Singular Value Decomposition:**  
 $A = U \Sigma V^T$   
 $U$ : orthogonal  $m \times m$  matrix with left singular vectors as columns  
 $V$ : orthogonal  $n \times n$  matrix with right singular vectors as columns  
 $\Sigma$ : "symmetric"  $m \times n$  matrix with  $\Sigma_{ii} = \sigma_i$  and 0 if  $i > j$   
 ( $\sigma_i$  are the singular values of  $A$  in descending order:  $\sigma_1 \geq \dots \geq \sigma_{\min(m,n)}$ )  
 (Remark 8.12)

If  $A$  has rank  $r$ :  $A = U_r \Sigma_r V_r^T$ , where  $U_r \in \mathbb{R}^{m \times r}$  is the first  $r$  left singular vectors,  $V_r \in \mathbb{R}^{n \times r}$  the first  $r$  right singular values and  $\Sigma_r \in \mathbb{R}^{r \times r}$  first  $r$   $\sigma_i$

$A A^T = U (\Sigma \Sigma^T) U^T$ , thus the left singular vectors of  $A$  are the eigenvectors of  $A A^T$  and the singular values are the square-root of the eigenvalues of  $A A^T$  (if  $m > n$ , the "missing" eigen values are 0)  
 Analogously:  $A^T A = V (\Sigma^T \Sigma) V^T$ , thus the right singular vectors of  $A$  are the eigenvectors of  $A^T A$  and the singular values are the square-root of the eigenvalues of  $A^T A$  (if  $m < n$ , the "missing" eigen values are 0) (Remark 8.1.3)

Direct consequence of SVD:  $A \in \mathbb{R}^{m \times n}$  with rank  $r$ :  $A = \sum_{k=1}^r \sigma_k u_k v_k^T$  (P8.1.4)  
 can be used for image compression

Every  $A \in \mathbb{R}^{m \times n}$  has an SVD (T8.1.5)

**Vector and matrix norms:**  
 (44)  $\|x\|_p = \left( \sum_{i=1}^n |x_i|^p \right)^{1/p}$   
 for  $p < \infty$ , and  $\|x\|_\infty = \max_i |x_i|$ . Notice that  $\|\cdot\|_2$  corresponds to the Euclidean norm that we have used in this course.  
 Definition 8.2.1 (Two matrix norms). Given a matrix  $A \in \mathbb{R}^{m \times n}$  we define two matrix norms:  
 •  $\|A\|_F$ , known as the Frobenius norm, is defined as  $\|A\|_F = \left( \sum_{i,j} A_{ij}^2 \right)^{1/2}$   
 •  $\|A\|_{op}$ , known as operator or spectral norm, is defined as  $\|A\|_{op} = \max_{\|x\|_2=1} \|Ax\|_2$

**Farkas Lemma:**  
 Definition 5.5.1. Let  $A \in \mathbb{Q}^{m \times n}$ ,  $b \in \mathbb{Q}^m$  and  $P = \{x \in \mathbb{R}^n \mid Ax \leq b\}$ .  $P$  is called a polyhedron. Let  $I = \{1, \dots, s\}$ . The projection of  $P$  on the subspace  $\mathbb{R}^I$  associated with the variables in the subset  $I$  is  $\text{proj}_I(P) = \{x \in \mathbb{R}^I \mid \exists y \in \mathbb{R}^{n-I} \text{ such that } (x, y) \in P\}$ .  
 Proposition 5.5.2.  $P \neq \emptyset$  iff  $1 \leq i \leq m$  and  $0 \leq y_i \leq m-1$  and  $0 \leq y_i^2 \leq b$  for all  $y_i \geq 0$  such that  $y^T A = 0$ .  
 Step 1 Partition the indices  $M = \{1, \dots, m\}$  of the rows of the matrix  $A$  into three subsets  $M_0 = \{i \in M \mid a_{ii} = 0\}$ ,  $M_1 = \{i \in M \mid a_{ii} > 0\}$  and  $M_2 = \{i \in M \mid a_{ii} < 0\}$ .  
 Step 2 - For every row with index  $i \in M_0$ , multiply the corresponding constraint by  $\frac{1}{a_{ii}}$ . This gives a new representation of the  $i$ -th row as follows  $a_{ii} \leq b_i + \sum_{j \in M_1} f_j$  for  $i \in M_0$ , where  $f_j = \frac{a_{ij}}{a_{ii}}$ ,  $f_j = -\frac{a_{ij}}{a_{ii}}$ .  
 - Every row with index  $i \in M_1$  can be rewritten as  $0 \leq b_i + \sum_{j \in M_1} f_j$  for  $i \in M_1$ , where  $f_j = b_j$ ,  $f_j = -a_{ij}$ .  
 - For every row with index  $i \in M_2$ , multiply the corresponding constraint by  $\frac{1}{a_{ii}}$ . This gives a new representation of the  $i$ -th row as follows  $a_{ii} \geq b_i + \sum_{j \in M_1} f_j$  for  $i \in M_2$ , where  $f_j = \frac{a_{ij}}{a_{ii}}$ ,  $f_j = -\frac{a_{ij}}{a_{ii}}$ .

**Step 3 Return:**  
 $Q = \{x \in \mathbb{R}^n \mid 0 \leq A_i + f_i \text{ for all } i \in M_0, A_i + f_i^2 \leq b_i + f_i^2 \text{ for all } i \in M_1, i \in M_2\}$ .  
 Theorem 5.5.3. The set  $Q$  returned in Step 3 is a polyhedron. Moreover,  $Q = \text{proj}_I(P)$ , where  $I = \{1, \dots, n-1\}$ .  
 Lemma 5.5.4. Let  $A \in \mathbb{Q}^{m \times n}$ ,  $b \in \mathbb{Q}^m$  and  $P = \{x \in \mathbb{R}^n \mid Ax \leq b\}$ . Let  $I = \{1, \dots, n-1\}$  and  $S_I = \{1, \dots, m-1\}$ . Then  $\text{proj}_I(P) = \text{proj}_{S_I}(\text{proj}_{S_I}(P))$ .  
 Definition 5.5.5. Let  $A \in \mathbb{Q}^{m \times n}$ ,  $b \in \mathbb{Q}^m$  and  $P = \{x \in \mathbb{R}^n \mid Ax \leq b\}$ . For  $k \in \{1, \dots, n\}$  let  $A^{(k)}$  be the submatrix of  $A$  with column vectors  $A_{\cdot i}$  for  $i \neq k$ . Let  $P^{(k)} = P$  and  $C^{(k)} = \mathbb{R}^n$ . Define for  $i \in \{1, \dots, n\}$   $C^{(i)} = \{x \in \mathbb{R}^n \mid x^T A_{\cdot i} = 0 \text{ for all } i = n-i+1, \dots, n\}$   
 $P^{(i)} = \{x \in \mathbb{R}^n \mid x^T A_{\cdot i} \leq b_i \text{ for all } i \in C^{(i)}\}$ .  
 Theorem 5.5.6.  $\text{proj}_{S_I}(P) = P^{(n)}$ .  
 Proof. We first verify that  $\text{proj}_{S_I}(P) \subseteq P^{(n)}$ . Indeed, take any  $x \in \text{proj}_{S_I}(P)$ . By definition there exists a real vector  $y$  of dimension 1 such that  $(x, y) \in P$ . Hence,  $(x, y)$  satisfies the following inequalities  $\sum_{i=1}^n A_{i1} x_i + A_{i2} y \leq b_i$  for all  $i \in \{1, \dots, n\}$ . This implies that for all  $y \in C^{(n)}$  we obtain that  $\sum_{i=1}^n A_{i1} x_i + \sum_{i=2}^n A_{i1} x_i = \sum_{i=1}^n A_{i1} x_i = x^T A_{\cdot 1} = y^T A_{\cdot 1} \leq b_1$  for all  $y \in C^{(n)}$ .  
 For the converse direction, we apply induction. Let  $i = 1$ . We have  $C^{(1)} = \{y \in \mathbb{R}^n \mid y^T A_{\cdot 1} = 0\}$  and  $P^{(1)} = \{x \in \mathbb{R}^n \mid x^T A_{\cdot 1} \leq b_1 \text{ for all } y \in C^{(1)}\}$ .  
 Our task is to show that  $P^{(1)} \subseteq \text{proj}_{S_I}(P)$ . This follows from Theorem 5.5.3. To see this, let  $x$  be the  $i$ -th unit vector in  $\mathbb{R}^n$ . Then  $x$  is the unit vector  $e_1$  for  $i = 1$ . Then  $e_1 \in C^{(1)}$ . Moreover, pick two indices  $i \in M_1$  and  $i \in M_2$ . Then  $\frac{1}{a_{ii}} e_i + \frac{1}{a_{ii}} e_i \in C^{(1)}$ .  
 This gives precisely the description of  $Q$  in Theorem 5.5.3.  
 This argument can be adapted to show the inductive step. The details are left as an exercise.  $\square$   
 Theorem 5.5.7 (The Farkas Lemma). Let  $A \in \mathbb{Q}^{m \times n}$ ,  $b \in \mathbb{Q}^m$ . Either there exists a vector  $x \in \mathbb{R}^n$  such that  $Ax \leq b$  or there exists a vector  $y \in \mathbb{R}^m$  such that  $y \geq 0$ ,  $y^T A = 0$  and  $y^T b < 0$ .  
 Proof. We refer to the notation introduced in Definition 5.5.5.  
 $C^{(n)} = \{x \in \mathbb{R}^n \mid x^T A_{\cdot i} = 0 \text{ for all } i = 1, \dots, n\} = \{x \geq 0 \mid x^T A = 0\}$ .  
 $P^{(n)} = \{0 \leq x^T b \text{ for all } x \in C^{(n)}\}$ . We conclude with Proposition 5.5.2 that  $P \neq \emptyset$  iff  $P^{(1)} \neq \emptyset$  iff  $\dots$  iff  $P^{(n)} \neq \emptyset$  iff for all  $y \geq 0$  with  $y^T A = 0$  we have  $y^T b \geq 0$ .  
 This leads to the following conclusion. Either  $P \neq \emptyset$  or  $P = \emptyset$ . This is equivalent to saying: either there exists a vector  $x \in \mathbb{R}^n$  such that  $Ax \leq b$  or there exists a vector  $y \in \mathbb{R}^m$  such that  $y \geq 0$ ,  $y^T A = 0$  and  $y^T b < 0$ .  $\square$

**Some more Matrices:**  
 Rotation matrix:  $R_\theta = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$   $R_\theta$  orthogonal.  
 Reflection matrix:  $A = I - 2vv^T$  reflects along Hypo-plane  $H = \{x \mid v \cdot x = 0\}$   
 Shear:  $\square \rightarrow \square \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$  Shear  $\square \rightarrow \square \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$   
 Rotation:  $\square \rightarrow \diamond \begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix}$  Reflection:  $\square \rightarrow \square \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

**Usefull formulas:**  
 $-(AB)^T = B^T A^T$  (L2.19)  
 $-(AB)^{-1} = B^{-1} A^{-1}$  (L3.9)  
 $-(A^T)^{-1} = (A^{-1})^T$  (L3.10)  
 - for permutation matrix  $P$ :  $P^{-1} = P^T$  (L3.16)  
 $\det(P) = \text{sgn}(P)$  (PG.0.7)  
 $\det(PA) = \det(A)$  (PG.0.21)  
 $-\text{rank}(A) = \text{rank}(A^T)$  (T4.29)  
 - for invertible  $m \times m$   $M$ :  $R(A) = R(MA)$  (L4.27)  
 $N(A) = N(MA)$  (L4.33)  
 $-N(A) = N(A^T A)$  (L5.1.11)  
 $-C(A^T) = C(A^T A)$  (L5.1.11)  
 - for orthogonal  $Q$ :  $Q^T Q = I$  (D5.43)  
 $Q Q^T = I$  (D5.43)  
 $Q^{-1} = Q^T$  (D5.43)  
 $C(A) = C(Q)$   
 $\det(Q) = 1 \text{ or } -1$  (PG.0.10)  
 $-A A^+ A = A$  (T5.5.11)  
 $-A^+ A A^+ = A^+$  (T5.5.11)  
 $-(A^T)^+ = (A^+)^T$  (T5.5.11)  
 - for triangular matrices  $T$ :  $\det(T) = \prod_{i=1}^n T_{ii}$  (PG.0.8)  
 $-\det(A^T) = \det(A)$  (T6.0.9)  
 $-\det(AB) = \det(A) \det(B)$  (PG.0.12)  
 $-\det(A^{-1}) = \frac{1}{\det(A)}$  (PG.0.13)  
 $-\text{Tr}(AB) = \text{Tr}(BA)$  (L7.1.14)  
 $-\text{Tr}(ABC) = \text{Tr}(BCA) = \text{Tr}(CAB)$  (L7.1.14)  
 - for symmetric matrices  $S$ :  $S = S^T$



### Multiple choice

If  $A^2$  is defined, then  $A$  must be square.

If  $A^2 = I$ , then  $A = I$ .

If  $A^2 = 0$ , then  $A = 0$ .

If  $A = \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix}$ , then  $A^n = \begin{bmatrix} 1 & na \\ 0 & 1 \end{bmatrix}$  for all  $n \in \mathbb{N}$ .

If  $AB = B$  for some choice of  $B$ , then  $A = I$ .

If both products  $AB$  and  $BA$  are defined, then  $A$  and  $B$  must be square.

If two columns of  $A$  are equal and  $AB$  is defined, the corresponding columns of  $AB$  must also be equal.

If two columns of  $B$  are equal and  $AB$  is defined, the corresponding columns of  $AB$  must also be equal.

If two rows of  $A$  are equal and  $AB$  is defined, the corresponding rows of  $AB$  must also be equal.

If two rows of  $B$  are equal and  $AB$  is defined, the corresponding rows of  $AB$  must also be equal.

If  $A$  and  $B$  are symmetric matrices and  $AB$  is defined,  $AB$  is also symmetric.

$N(A) = N(2A)$   $C(A) = C(2A)$

Let  $U_1, U_2$  be arbitrary subspaces of  $\mathbb{R}^n$ . Which of the following subsets of  $\mathbb{R}^n$  must be subspaces of  $\mathbb{R}^n$  as well?

$U_1 \cap U_2 = \{0\}$   $U_1 + U_2 = \{u_1 + u_2 \mid u_1 \in U_1, u_2 \in U_2\}$

### General

If the solutions of an SLE are  $x_1 = 0, x_2 = 0, x_3 = 1$ , the system has infinite many solutions.

Let  $A$  be a real  $2 \times 4$  matrix with rank 2. Then the SLE  $Ax = b$  has a non-trivial solution.

If  $A$  is invertible it holds:  $ABA^{-1} = B$

$\mathbb{R}^{n \times n} \rightarrow \mathbb{R}, A \mapsto \text{trace}(A)$  is linear

$\mathbb{R}^{n \times n} \rightarrow \mathbb{R}, A \mapsto \det(A)$  is linear

Let  $D \in \mathbb{R}^{2 \times 2}$ ,  $\dim(\text{Ker } D) = 2$  only if  $D = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

If  $A^2$  is invertible, so is  $A^3$

$\det(A^2) \neq 0 \Leftrightarrow \det(A) \neq 0 \Leftrightarrow \det(A^3) \neq 0$

If  $A$  is regular and  $A^2 = A$ , then  $A = I$

For linear dependent  $x, y, z$  it holds  $x = \alpha y + \beta z$ . Only  $\alpha, \beta$  can be dependent

If  $A$  and  $A^2 \in \mathbb{R}^{n \times n}$  and  $A^2$  is regular,  $A^3$  is invertible

$\forall x \in \mathbb{R}^n, \|Ax\|_2 \leq \|A\|_2 \|x\|_2$

Let  $f: \mathbb{R}^n \rightarrow \mathbb{R}, f(x) := \|Ax\|_2$ . The function  $f$  is a norm in  $\mathbb{R}^n$

$\|AB\|_2 \leq \|A\|_2$

Given are the orthogonal matrices  $A$  and  $B$  with the same dimension. Which of the following properties is true?

The matrix product  $AB$  is orthogonal, but  $BA$  is not orthogonal.

The matrix product  $BA$  is orthogonal, but  $AB$  is not orthogonal.

The matrix product  $AB$  and the matrix product  $BA$  are orthogonal.

The matrix product  $AB$  and the matrix product  $BA$  are not orthogonal. It holds that  $A^T = A^{-1}$ ,  $B^T = B^{-1}$  and further  $(AB)^T = B^T A^T = B^{-1} A^{-1} = (AB)^{-1}$  and also vice versa  $(BA)^T = (BA)^{-1}$

We have  $\mathbb{R}^n$  with the standard scalar product  $\langle \cdot, \cdot \rangle$  and 2-norm. Let  $A$  be a real  $n \times n$  matrix. Which statements are correct?

$\forall x, y \in \mathbb{R}^n$  it holds that  $\langle Ax, Ay \rangle = \langle Ax, y \rangle$

$A^T = A^{-1} \Leftrightarrow \forall v, w \in \mathbb{R}^n \langle Av, Aw \rangle = \langle v, w \rangle$

$A^T = A^{-1} \Leftrightarrow \forall v \in \mathbb{R}^n \langle Av, Av \rangle = \|v\|_2^2$

Let  $B$  be another real  $n \times n$  matrix.  $A^T = A^{-1}$  and  $B^T = B^{-1} \Rightarrow$  the inverse of  $AB$  exists and it is:  $(AB)^{-1} = (AB)^T$

Given are orthogonal matrices  $A \in \mathbb{R}^{n \times n}$  and  $B \in \mathbb{R}^{m \times m}$ . Which of the following statements are correct?

The matrix  $A^T$  is orthogonal

The matrix  $A + B$  is orthogonal

The matrix  $A + A^T$  is orthogonal

The matrix  $AB^{-1}$  is orthogonal

Given is a lower triangular matrix  $A \in \mathbb{R}^{3 \times 3}$  whose entries are non-negative integers and where entries which occur only once or are equal to zero. Which of the following options are possible for the value of the determinant  $\det(A)$ ?

3

5

7

-2

35

$A$  is triangular  $\det(A) = a_{11} \cdot a_{22} \cdot a_{33} = 0 \Rightarrow \det(A) = 0$  or  $\det(A) = a_{11} \cdot a_{22} \cdot a_{33}$

### Rank

Let  $B \in \mathbb{R}^{3 \times 1}$  and  $C \in \mathbb{R}^{2 \times 3}$ .  $BC$  can have rank 3.

### Vectorspaces

Let  $V$  be a vector space over  $\mathbb{R}$  with scalar product  $\langle \cdot, \cdot \rangle$  and let  $F: V \rightarrow V$  be a linear map. If it holds that  $\forall v \in V, \langle v, F(v) \rangle = 0$ , then  $F$  is necessarily the null map, i.e.,  $F(v) = 0$  for all  $v \in V$ .

Given a vector space  $V$  with a norm  $\|\cdot\|$ . For all  $u, v \in V$ , we have  $\|u\| \leq \|u+v\|$ .

counterexample:  $u = -v = (1, 1)^T$

$\mathbb{R} \subset V$  and  $W$  be a subspace of  $V$ .  $\mathbb{R} \subset W = \text{span}\{S\} \subset W$

In a vector space of finite dimension with scalar product, one can complete any set of orthonormal vectors to form an orthonormal basis.

A vector space of finite dimension with scalar product has an orthonormal basis. Corollary 6.7

Consider the vector space  $\mathbb{R}^n$  with the Euclidean scalar product. The scalar product of two unit vectors can be arbitrarily large. Cauchy-Schwarz:  $|\langle u, v \rangle| \leq \|u\| \|v\|$  ( $\|u\| = \|v\| = 1 \Rightarrow |\langle u, v \rangle| \leq 1$ )

We again consider  $\mathbb{R}^n$  with the Euclidean scalar product. Can we find any number of pairwise orthogonal unit vectors in this vector space?

Let  $V$  the standard vector space of all  $2 \times 2$  matrices. Which of the following are subspaces of  $V$  ( $B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ )

$\{A \in V \mid A \text{ is invertible}\}$

$\{A \in V \mid A^2 = 0\}$

$\{A \in V \mid A^T = A\}$

$\{A \in V \mid A^T B = BA\}$

Consider the vector space  $F$  of functions of  $\mathbb{R} \rightarrow \mathbb{R}$  with the operations addition  $(f+g)(x) = f(x) + g(x)$  and scalar multiplication  $(\lambda f)(x) = \lambda f(x)$ . This includes the subspace  $P_2 = \{a_0 + a_1 x + a_2 x^2 \mid a_i \in \mathbb{R}\}$  of polynomials of degree  $\leq 2$ . Which of the following statements are correct?

$\text{Span}\{x+1, x^2+1, x^2+1\}$  is equal to  $P_2$ .

$x+1, x^2+1, x^2+1 \in P_2$  are linearly independent

$x+1, x^2+1, x^2+1 \in P_2$  form a generating set (spanning set) of  $P_2$ .

$x+1, x^2+1, x^2+1 \in P_2$  form a generating set (spanning set) of  $P_2$ .

The polynomials  $x+1, x^2+1, x^2+1 \in P_2$  form a basis of  $P_2$ .

Let  $V, W$  be finite dimensional vector spaces over a space  $K$ . Let  $F: V \rightarrow W$  be a linear mapping and  $\{v_1, \dots, v_n\}$  a basis of  $V$ . Then it holds that:

$F(v_1), \dots, F(v_n)$  are linearly independent if  $F$  is surjective.

$F(v_1), \dots, F(v_n)$  are linearly independent if  $F$  is injective.

$F(v_1), \dots, F(v_n)$  form a generating end system if  $F$  is surjective.

$F(v_1), \dots, F(v_n)$  form a generating end system if  $F$  is injective.

$F(v_1), \dots, F(v_n)$  form a basis if and only if  $F$  is an isomorphism.

Let  $V, W$  be two real vector spaces with scalar products, let  $B$  be an orthonormal basis of  $V$  and let  $F: V \rightarrow W$  be an orthogonal mapping. Which of the following statements is true?

$F$  is an isomorphism.  $F$  is only an isomorphism if  $\dim V = \dim W < \infty$ .

$\|F(v)\|_W = \|v\|_V$  for all  $v \in V$ . Follows directly from the definition of the scalar product induced norm and the orthogonality of  $F$ .

If  $\dim V, \dim W < \infty$ , it is possible that  $\dim V > \dim W$ . With  $\dim V > \dim W$  there is no orthogonal mapping  $F: V \rightarrow W$ .

$F$  is not injective.

$F$  is angle-preserving, i.e., for all  $v, w \in V$  it holds that  $\langle F(v), F(w) \rangle = \langle v, w \rangle$ .

$F$  is an isomorphism to the image of  $F$ .  $F$  is injective as shown before and obviously surjective to its image. Since the Inverses of linear mappings are also linear,  $F$  is therefore an isomorphism.

The set of images  $F(B)$  is an orthonormal basis of  $W$ . Since  $F$  is not necessarily an isomorphism, the image space can be smaller than  $W$ .

The set of images  $F(B)$  is an orthonormal basis of  $\text{Im}(F)$ . This follows from the isomorphism property of  $F$  from the question above.

If it exists,  $F^{-1}: \text{Im}(F) \rightarrow V$  is orthogonal. It exists as seen above. The orthogonality of the

inverse follows directly from the definition of orthogonal mappings.

### Det

Let  $Q$  be unitary and  $A \in \mathbb{R}^{n \times n}$ .  $\det(QA) = \det(A)$   $\det(Q) = \pm 1$

If  $A, B, P \in \mathbb{R}^{n \times n}$  and  $P$  is invertible with  $A = PBP^{-1}$  then:  $\det(A) = \det(B)$

Given is a matrix  $A \in \mathbb{R}^{n \times n}$  with entries  $a_{ij} = ij$  and  $n > 1$ . Which statement is correct?

$\det(A) = 1$

$\det(A) = 0$

$\det(A) = (-1)^n$

$\det(A) = (-2)^n$

Which of the following statements are not correct for arbitrary  $n \times n$ -matrices  $A$  and  $B$ ?

$\det(A+B) = \det(A) + \det(B)$

$\det(AB) = \det(BA)$

If  $A$  is singular then  $AB$  is also singular

$\det(AA^T A) = (\det(A))^3$

Let  $A, B \in \mathbb{R}^{n \times n}$  with  $AB = -BA$

$\det(AB) = \det(-BA)$

$\det(A)\det(B) = -\det(A)\det(B)$   $n$  has to be even  $\Rightarrow n \in 2\mathbb{N}$

Either  $A$  or  $B$  has a zero-determinant

$A$  and  $B$  have to be singular

$ABx = 0$  has more than one solution (Lösungsschar)

$ABx = c$  can have no, one and  $\infty$  many solutions, if  $c \in \mathbb{R}^n, c \neq 0$ .

It has to be  $A = 0$  or  $B = 0$ .

Which of the following statements are correct for an arbitrary  $n \times n$ -matrix  $A$  and for arbitrary  $n$ ?

$\det(2A) = 2\det(A)$

$\det(-A) = \det(A)$

$\det(A^2) = \det(A)^2$

Let  $A$  be a triangular matrix with the property  $a_{ij} = 0$  for  $i+j > n+1$  (so there are zeros at the bottom right). The determinant can be calculated using the formula  $\det(A) = a_{11} \cdot a_{22} \cdot a_{33} \cdot \dots \cdot a_{nn}$ .

Let  $A, P, Q \in \mathbb{R}^{n \times n}$  where  $P$  is permutation matrix and  $Q$  is a unitary matrix.

$\det(PA) = \det(A)$

$\det(PAP) = \det(A)$

$\det(QA) = \det(A)$

### Eigenvalue/Eigenvectors

with  $A_A(\lambda) = (\lambda - 1)^3 + 3 \Rightarrow A \in \mathbb{R}^{3 \times 3}$  invertible it isn't eigenvalue  $\Rightarrow A$  is regular

Let  $x_1$  and  $x_2$  be eigenvectors of  $A$ , so is  $x_1 + x_2$  an eigenvector

If  $A \in \mathbb{R}^{n \times n}$  and  $A_A(\lambda) = (\lambda - 1)^n + 2$ ,  $A$  is invertible

Similar matrices have the same eigenvalues

Similar matrices have the same eigenvectors

Every  $n \times n$  matrix has linear independent eigenvectors

Eigenvectors, which correspond to the same eigenvalue are always linear dependent

If a real matrix has a eigenvector, it follows that the matrix has infinitely many eigenvalues

Every rotation in  $\mathbb{R}^2$  has the eigenvalue  $\lambda = 1$

If  $\lambda_1$  with  $v$  and  $\lambda_2$  with  $w$ , so is  $(\lambda_1 + \lambda_2)$  a eigenvalue with eigenvector  $v + w$

Let  $A \in \mathbb{R}^{3 \times 3}$  with eigenvalues  $\lambda_1, \lambda_2, \lambda_3$

$A$  is diagonalizable if all eigenvalues are different

If  $A$  is diagonalizable, all eigenvalues have to be different

$A$  is diagonalizable if it has 3 eigenvectors. They have to be linearly independent

If  $\lambda_1 = 2, \lambda_2 = -1, \lambda_3 = 1$  and  $B = A^3 - 3A^2$  then is  $B$  diagonalizable

If  $AP = PD$  and  $D$  is a diagonal matrix then the columns of  $P$  are eigenvectors of  $A$ . Only if the eigenvalues of  $A$  are on the diagonal of  $D$ .

Let  $A \in \mathbb{R}^{n \times n}$  be positive definite and symmetric. Further, let  $\lambda_1, \dots, \lambda_n$  be the eigenvalues to the eigenvectors  $v_1, \dots, v_n$

$A^2$  has at least one eigenvalue with a strictly positive imaginary part

It holds that  $\lambda_j > 0$  for all  $j = 1, \dots, n$

$A$  has at least one eigenvalue which satisfies:  $\text{geom. mult.} < \text{alg. mult.}$

The eigenvalues are pairwise distinct:  $\lambda_j \neq \lambda_i$  if  $j \neq i$

There exist positive real numbers  $\alpha > 0$  such that  $\alpha^T A x \geq \alpha^T x$  for all  $x \in \mathbb{R}^n$

Let  $A \in \mathbb{R}^{2 \times 2}$  with  $\text{Rank}(A) = 1$  and  $\text{Trace}(A) = 2$ . What are the eigenvalues?

1 is an eigenvalue

0 is an eigenvalue

2 is an eigenvalue

-5 is an eigenvalue

5 is an eigenvalue. Since  $\det(A) = 0$  and  $\text{Trace}(A) = 2 \Rightarrow 0 = \lambda_1 \cdot \lambda_2$  and  $2 = \lambda_1 + \lambda_2$

### Decompositions

A matrix  $A \in \mathbb{R}^{n \times n}$  with  $n$  eigenvalues has  $2^n$  normal spectral decompositions since the sign can be changed at times

Let  $A \in \mathbb{R}^{n \times n}, n \geq n$  be a matrix with rank  $k$ . Denote the QR-decomposition of  $A$  as  $A = QR$ , where  $Q \in \mathbb{R}^{n \times n}$  has orthonormal columns, and  $R \in \mathbb{R}^{n \times n}$  is an upper (right) triangular matrix. Which one of the following statements is always true?

$\text{Rank } A < \text{Rank } R$

$QQ^T = I$

If  $A$  has linearly independent columns, we have  $\text{Rank } R = m$

If  $A$  has linearly independent columns, we have  $\text{Rank } R = n$

Let  $A \in \mathbb{R}^{m \times n}$  with linearly independent columns and  $A = Q_1 R_1 = Q_2 R_2$ , two QR-Decompositions of  $A$

$Q_1^T Q_2$  is orthogonal

$Q_1^T Q_2$  is a upper and lower triangular matrix and therefore a diagonal matrix

$Q_1^T Q_2 = I$

$\text{Rank}(R_1) = m$

$\text{Rank}(R_1) = n$

$\text{Rank}(R_2) = m$

$\text{Rank}(R_2) = n$

$Q_1^T Q_2$  is regular

Let  $A, B \in \mathbb{R}^{n \times n}$ ,  $B$  is regular and  $B = QR$

$f: \mathbb{R}^n \rightarrow \mathbb{R}, x \mapsto \|Ax\|_2$ , is a norm in  $\mathbb{R}^n$

If  $B$ , such that  $A^k$  is invertible, so is  $A$  not invertible

$\forall x \in \mathbb{R}^n, \|Ax\|_2 \leq \|A\|_2 \cdot \|x\|_2$

$\det(B) = \det(A)$

$\|B\|_2 = \|A\|_2$

$\phi: \mathbb{R}^n \rightarrow \mathbb{R}, x \mapsto \|Qx\|_2$ , is a norm in  $\mathbb{R}^n$

$AB$  is regular, but  $BA$  is not necessarily

$\|AB\|_2 \leq \|A\|_2$

### Basis

The transformation matrix of a basis transformation between orthonormal bases is the identity matrix. The transformation matrix of base transformation between orthonormal bases is orthogonal, the identity matrix is only one possibility.

The inverse of the transformation matrix of a base transformation between orthonormal bases is its Hermitian transpose.

$A \in \mathbb{R}^{n \times n}$  is an orthogonal matrix if and only if its columns form an orthonormal basis of  $\mathbb{R}^n$  with respect to the Euclidean scalar product.

The change of basis matrix is unitary (if  $\mathbb{R} = \mathbb{C}$ ) or orthogonal (if  $\mathbb{R} = \mathbb{R}$ ) if both bases are orthonormal.

### Procedures

The Gram-Schmidt orthogonalisation method can be used to compute an equally large set of linearly independent vectors from a set of linearly dependent vectors.

Let  $x_1, \dots, x_n \in \mathbb{R}^n$  be a set of  $n$  vectors. Using the Gram-Schmidt process, we can always produce a unit-length and pairwise orthogonal vectors. Gram-Schmidt needs linear independent vectors

Let  $A \in \mathbb{R}^{n \times n}, n < \infty$ . Let  $Ax = b$  be a system of linear equations and let  $x$  be a solution in the least squares sense. Which statement is always correct?

The vector  $(\delta Ax)$  is orthogonal to the row space of  $A$ .

The vector  $(\delta Ax)$  is orthogonal to the column space of  $A$ ,  $\in$  normal equations

$ax$  is the null space of  $A$ .

The solution  $x$  does not always exist

### Kernel/Image

If the nullspace of an  $8 \times 7$  matrix is 5-dimensional, the row space has dimension 3  $n = \dim(\text{Ker}(A)) = 7 - 5 = 2 = \dim(\text{Im}(A))$



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$